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NPS/003/005 – Technical Specification for LV Fuseboards, Fusepillars, Fusecabinets and Circuit Breakers.

1. Purpose

The purpose of this document is to detail the technical requirements for fuse boards, fuse pillars, fuse cabinets and circuit breakers for use on CE Electric UK's low voltage distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NPS/003/005	2.0	Technical Specification for LV Fuseboards, Fuse pillars, Fuse cabinets And LV CBs.

2. Scope

This document applies to all fuseboards, fusepillars, fusecabinets and circuit breakers used to supply, control and protect low voltage (lv) cable connected distribution circuits on CE Electric UK's network. National and international standards have been specified where applicable. See Section 5 for Definitions.

Appendix 3 contains self certification conformance declarations, to be completed by the supplier for each variant offered.

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3. Technical Requirements

3.1 General

All equipment covered by this specification shall be in accordance with the latest versions of ENATS 37-2 and BS EN 60439 - 5. This specification supplements, or varies, ENATS 37-2 and BS EN 60439 - 5 only where necessary to ensure that the equipment is fit for purpose for use on CE Electric UK's distribution network.

3.2 Insulation

All fuse boards, fuse pillars, fuse cabinets and circuit breakers (CB's) shall be insulated, shielded type units designed to prevent inadvertent access to live parts.

3.3 Connection of Temporary Flexible Leads

- 3.3.1 Fuseboards, fusepillars and fusecabinets shall be equipped to allow the connection of external devices, such as mobile generators, via temporary leads equipped with Litton VEAM type connectors, where required.
- 3.3.2 An option shall be available to allow the door(s) of the fuse board, fuse pillar, fuse cabinet or CB to be closed and locked whilst generator leads are connected. This shall not affect the degree of protection offered (IP33 according to BS EN 60529) or vermin protection (IP3X).

3.4 Measurement of Current and Voltage

- 3.4.1 Fuseboards, fusepillars, fusecabinets shall include facilities for generators to be synchronised and for electrical testing across the transformer LV links, fused connections, utilising red spot fuses, shall be made on all three phases, on both sides of the transformer disconnecter (i.e. a total of 6 connections).

These shall be terminated on 4mm banana plug test sockets. The areas around these test sockets shall be suitably insulated for a minimum distance of 13mm from the outside edge of the socket in order to reduce the risk of a short circuit during lead insertion/removal, connection of crocodile clips, etc.

- 3.4.2 To allow current measurements to be made on each phase of each feeder the unit shall allow for the safe and practical use of a clip-on ammeter.
- 3.4.3 Facilities to test all incomers, busbars, and outgoing ways shall be provided. These facilities shall allow for the use of test lamps/probes with characteristics in accordance with ENA Engineering Recommendation M15/4. ER M15/4 includes stipulation for test probe dimensions of 4mm max diameter and 15mm reach.

3.5 Bulk LV Supplies

Customers requiring LV supplies in the range 330 – 1000kVA will normally be provided by low voltage circuit breakers (LV CB's) integrated in a fuseboard, fusepillar or fusecabinet.

- 3.5.1 LV CBs shall be fitted with remote emergency shunt trip facility that will accept inputs of either 30V DC or 110V AC.
- 3.5.2 LV CB's shall have overlapping ranges of protection and shall be rated at 800A, 1600A or 2500A.
- 3.5.3 The LV CB shall be fully rated and tested as a point of isolation.
- 3.5.4 The LV CB arrangement shall incorporate metering CT's.

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- 3.5.5 Metering secondary wiring shall be marshalled in a compartment that is accessible without approach to bare primary conductors and without access into the main LV fuseboard. Within this compartment the wiring and terminal blocks shall comply fully with ENATS 50-18 and ENATS 50-19.
- 3.5.6 Fuseboards, fusepillars and fusecabinets fitted with LV CB's shall also be equipped with a minimum of two cable feeder ways in addition to the LV CB. A suitably rated transformer LV disconnect, that controls both the CB and the feeder ways, shall be provided.

3.6 Resistance to Physical Interference

- 3.6.1 Access to operational features and test points of fuseboards, fusepillars and fusecabinets shall be intruder resistant to an equivalent of Security Rating Classification 1 in accordance with the current version of BRE Certification Loss Prevention Standard LPS 1175.
- 3.6.2 Fuseboards, fusepillars and fusecabinets shall be capable of being upgraded to Security Rating Classification 2. This option shall be retrofittable on site and preferably without the need to make any part(s) of the equipment dead.

As a minimum, door(s) shall be upgradeable on site and without having to make any part(s) of the fuseboard dead.

3.7 Phase Reversal Option

Fuseboards, fusepillars and fusecabinets shall be capable of being supplied with or upgraded to accept a phase reversal kit. This kit is required to transpose red and blue phases on units to be used at NEDL 20kV substations, in order to provide standard system phasing of R-Y-B / L1-L2-L3 / B-B-G top to bottom on the LV busbars.

3.8 Fuse Ways / Carriers

Each phase of the fuse way shall be capable of independent operation. The fuse carriers shall be of a through grip shrouded type, manufactured from a suitable insulating material that provides a secure connection to the fuse link by means of wedge connections and insulated contact tightening thumbscrews. The design of the thumbscrews must ensure that when in position, the whole fuse assembly cannot be dismantled when the thumbscrews are released to their full extremities. The fuse carrier shall be suitable for accommodating fuse links with 82mm or 92mm fixing centres in accordance with BS 88 Part 5. Fuse carriers shall not be made of porcelain.

The following additional tests shall be carried out on the fuse carriers:-

The test shall be carried out on each design of fuse carrier including wedge mechanisms.

The test shall be made at an ambient temperature between 15°C and 25°C.

The test voltage of 3.75 kV (rms.) shall be applied for one minute between:

Both of the fuse carrier terminals without the fuse link being fitted.

One of the fuse carrier terminals and metal foil which is wrapped around surfaces of the insulating material of the fuse carrier, including thumbscrews, if any, which can be touched during live replacement or withdrawal of fuse carriers. For this test, a fuse link of the largest dimensions intended for the fuse carrier shall be fitted.

Following the high voltage test, the fuse carrier shall complete a humid atmosphere test as detailed in sub-clause 8.2.4.2 of BSEN 60269 – Part 1. The insulation resistance at the conclusion of the test shall be measured; the insulation resistance measured shall not be less than 5 Megohms.

Mechanical strength: The following additional test shall be carried out:

Torque tests on thumbscrews: The test shall be carried out on six random samples of the insulating thumbscrew, complete with wedge operating screw, to confirm mechanical strength. Each wedge

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operating screw shall in turn be placed in vice and a torque applied in a clockwise direction to the insulated thumbscrew using a torque spanner adapted as necessary. Each insulated thumbscrew shall withstand a torque of 12Nm without fracturing or turning on the shank of the wedge operating screw.

Verification of short circuit withstand strength: Short time withstand tests shall be under-taken with the minimum number of distributor units to be fitted in service and, in the case of short-circuit withstand ratings verified using prospective short circuit current, the supply voltage shall be equal to 1.1 times the rated operational voltage.

Fuse links are not to be supplied with fuse carriers

3.9 Door Stays

Fusepillars and fusecabinets shall be provided with door stays to allow the door to be held open in the 90°, 135° and 180° positions

3.10 Ancillary Equipment

Fuseboards, fusepillars and fusecabinets shall be capable of being provided with wiring suitable for the fitting of a voltage transformer.

3.11 Cable Types, Sizes and Terminations

Notwithstanding the requirements of ENATS 37-2, LV fuseboards, fusepillars, fusecabinets and CBs shall be designed to accommodate the following cable types:

Incoming Transformer Cables

For incoming transformer cables the design shall have 4 x 10mm fixing points, and provision for earthing the earth screens at the LV board and shall accommodate:

- a) 1c armoured stranded Cu 800mm² to BS 5467, Table 4.
- b) 1c sectoral Al XLPE insulated up to 480mm² to BS7889, Table 4.
- c) 3c 185mm² waveform cable to BS7870 3.4 (all cores connected together, and terminated in a single lug. An arrangement, when required should be available to terminate the screen wires to the neutral busbar.

Outgoing Feeder Way Cables

Phase conductors of outgoing cables: 185mm² or 300mm² 3c CNE or 4c SNE Waveform to BS7870 3.4 shall be connected by 1x16mm fixing points, with provision for earthing the cable screen wires.

3.12 Heater

An option for an unswitched metal clad heater, rated between 100W and 150W, shall be provided.

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3.13 Padlocking of Outgoing Distributor ways in the Open Position

Facilities for padlocking outgoing fuse ways in the open position shall be provided as standard.

3.14 Mechanical Cable Termination Clamps

The units shall be provided complete with mechanical shear bolt clamps for the terminating of the incoming and outgoing cables

4. References

4.1 External Documentation

Reference	Title
ENATS 37-2	Substation cable distribution boards
ENATS 50-18	Design and application of ancillary electrical equipment
ENATS 50-19	Standard numbering for small wiring (for switchgear and transformers together with their associated relay panels)
BS EN 60439 - 5	Low-voltage switchgear and controlgear assemblies. Particular requirements for assemblies for power distribution in public networks
ENA ER M15/4	Fused flexible leads for use with portable voltage measuring instruments at voltages up to 600V ac/dc
BRE LPS 1175	Building Research Establishment Certification Loss Prevention Standard LPS 1175 – Requirements and Testing Procedures for the LPCB Approval and Listing of Burglary Resistant Building Components, Strong points and Security Enclosures.
BS7870	LV and MV polymeric insulated cables for use by distribution and generation utilities.
BS7889	Electric cables. Thermosetting insulated, unarmoured cables for a voltage of 600/1000 V
BS5467	Electric cables. Thermosetting insulated, armoured cables for voltages of 600/1000 V and 1900/3300 V

4.2 Internal documentation

Reference	Title
NPS/003/005	Technical Specification for LV Fuseboards, Fuse pillars, Fuse cabinets And LV CBs.

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4.3 Amendments from Previous Version

Clause	Subject	Amendments
2.1	Contents	Table of contents added.
3.8	Fuse Carriers	Additional testing requirements.
3.11	Cable Type / Sizes	Amended to include 4c waveform (SNE) cable and use of 3c 185 waveform (cores "bunched" for transformer connections).
3.5	Bulk LV Supplies	Previous "large" lv supply defined.
4.1	External References	BS EN 60439-5 - title added
4.3	Amendments	Amendments table added.
5	Definitions	Definitions of Fuseboards, Fusepillars, Fusecabinets and Circuit Breakers added.

5. Definitions

Term	Definition
ENATS	Energy Networks Association Technical Specification
Fuseboard	Indoor Wall Mounted (ENATS 37-2 ref: SCDB-I)
Fusepillar	Free Standing Outdoor Ground Mounted (ENATS 37-2 ref: SCDB-CCO)
Fusecabinet	Outdoor Transformer Mounted (ENATS 37-2 ref: SCDB-TMO)
Circuit Breaker	Air Circuit Breaker (ACB) or Mould Case Circuit Breaker (MCCB)

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6. Authority for issue

6.1. Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
Vic Gorny	Standards Engineer	Vic Gorny	19/03/2010

6.2. Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	29/03/2010

6.3. Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
K Midgley	Standards Engineer	K Midgley	19/03/2010

6.4. DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	14/04/2010

6.5. Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	31/03/2010

6.6. Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	31/03/2010

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Appendix 1 - Addendum to Supplier Requirement

Items to be supplied with each Board/Pillar/Cabinet/CB

Copies of routine test results.

These shall include a listing of the results of micro-ohmmeter tests of the resistance between the incoming, transformer side, busbars and each of the outgoing contacts. This test shall be done at the end of all production procedures, but before any over voltage tests.

One copy to be supplied as weatherproof, durable, hard copy attached to the unit being supplied and one copy to be supplied electronically

Handling Instructions

A weatherproof, durable, permanently fixed label with lifting/slinging arrangements with and without packaging, shall be attached to the unit.

This must include gross weight, attachment points, recommended slinging arrangements and indication of centre of gravity.

If appropriate this label shall be repeated on the outside of the packaging.

Manufacturers may provide alternative tenders for items not complying with the above specification. This shall be clearly stated together with detailed descriptions of any variation from the specification, together with drawings and test results.

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from these standards or specifications.

Summary of Variants:

Transformer Mounted (close coupled)	-----	-----	All to be available as outdoor spec, with possible indoor option
Free Standing	-----	Both to be available with, or without, disconnecter (Tx) isolation links	
Wall Mounted	To be available as Bottom, Top, LHS or RHS entry		

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Schedules of Variants

4, 6 and 8 Way units may be accepted in place of 3, 5 and 7 Way units

OUTDOOR SPECIFICATION	3 way & Tx disconnect 800A busbars	5 way & Tx disconnect 1600A busbars	7 way & Tx disconnect 1600A busbars
Close coupled			
Free standing			
Wall mounted top entry			
Wall mounted RHS side entry			
Wall mounted LHS side entry			
Wall mounted bottom entry			
INDOOR SPECIFICATION	3 way & Tx disconnect 800A busbars	5 way & Tx disconnect 1600A busbars	7 way & Tx disconnect 1600A busbars
Wall mounted top entry			
Wall mounted RHS side entry			
Wall mounted LHS side entry			
Wall mounted bottom entry			

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Units with LV CBs	
LV CB 800A, close coupled, in standard cabinet, with Tx disconnecter and 2 LV distribution ways	
LV CB 800A, close coupled, in minimised cabinet, without Tx disconnecter or LV distribution ways	
LV CB 800A, free standing, in bottom entry & exit cabinet, without Tx disconnecter or LV distribution ways	
LV CB 800A, wall mounted, in bottom entry & exit cabinet, without Tx disconnecter or LV distribution ways	
LV CB 1600A, close coupled, in standard cabinet, with Tx disconnecter and 2 LV distribution ways	
LV CB 1600A, close coupled, in minimised cabinet, without Tx disconnecter or LV distribution ways	
LV CB 1600A, free standing, in bottom entry & exit cabinet, without Tx disconnecter or LV distribution ways	
LV CB 1600A, wall mounted, in bottom entry & exit cabinet, without Tx disconnecter or LV distribution ways	
LV CB 2500A, close coupled, in standard cabinet, with Tx disconnecter and 2 LV distribution ways	
LV CB 2500A, close coupled, in minimised cabinet, without Tx disconnecter or LV distribution ways	
LV CB 2500A, free standing, in bottom entry & exit cabinet, without Tx disconnecter or LV distribution ways	
LV CB 2500A, wall mounted, in bottom entry & exit cabinet, without Tx disconnecter or LV distribution ways	

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Optional Extras/Variations

Option/Variance	
Higher security door/features (state what this option consists of)	
Red-Blue phase reversal kit (state which LV Fuseboards, Fuse pillars, Fuse cabinets and CBs this is/isn't available for)	
Heater option (state rating proposed)	
Provision and wiring for a voltage transformer	
Arrangement to allow termination of 185mm ² waveform cable screen wires:- 1/ 4 cables (800A unit) 2/ 7 cables (1600A unit)	

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Appendix 2 - Logistical requirements

To enable CE Electric UK to store the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide details of the recommended storage environment with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to;

- | | | | |
|---|---------------------|---|-----------------------|
| * | Ambient temperature | * | Atmospheric corrosion |
| * | Humidity | * | Impact |
| * | Water | * | Vibration |
| * | Dust | * | Solar radiation |

The Tenderer shall ensure that each item is suitably packaged and protection to maintain the product and packaging as "fit for service" prior to installation taking account of the potential for an outdoor storage environment. All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub packages all the sub packages shall unless varied by this specification, be supplied securely packaged together. Where items are provided in bagged/boxed form the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. Tenderer shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product. Where the Tenderer is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple package of like products. Where products packages tendered are made up of sub packages each sub packages shall be marked. As a minimum requirement the following shall be included;

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * CE Electric UK product code
- * Weight

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Appendix 3 - SELF CERTIFICATION CONFORMANCE DECLARATION

All Fuseboards, Fusepillars, Fuse cabinets and Circuit Breakers (CB's) shall comply with the latest issues of the relevant national and international standards, including ENATS 37-2 and BS EN 60439 – 5. Additionally this technical specification is intended to amplify and/or clarify requirements relating to these Standards.

This self declaration sheet identifies the clauses of the aforementioned standards relevant to Fuseboards, Fusepillars, Fusecabinets and Circuit Breakers (CB's) for use on CE Electric UK's distribution network. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non- conformance shall be entered
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

**Details of the cable type(s)
suitable for installation:**

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each item or variant submitted.

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Appendix 4 Cont'**Technical Specification**

• <u>NPS/003/005</u>			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
3.1	Compliance with BS EN60439		
3.2	Minimum shielded categorisation		
3.3.1	Litton Veam type connectors		
3.3.2	Facility for closing door, with generator leads connected		
3.4.1	Six test sockets, fused and with insulated surrounds		
3.4.2	Usability of clip on ammeter/CT's		
3.4.3	Usability of test lamps		
3.5.1	Large LV supplies in standard cabinet,		
3.5.2	LV CB have 30V DC trip		
3.5.3	Minimum set of LV CBs with overlapping protection ranges.		
3.5.4	LV CB rated, and type tested, as point of isolation		

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<u>NPS/003/005</u>			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
3.5.5	LV CB arrangement incorporates metering CT's		
3.5.6	Metering secondary wiring in compliant compartment.		
3.5.7	Option of LV CB & feeder ways		
3.6.1	Equivalent to BRE LPS1175 Class 1		
3.6.2	Upgradeable to equivalent of security class 2		
3.7	R-B phase reversal kit option		
3.8	Non-porcelain carriers		
	Testing requirements		
3.9	Door stay		
3.10	Ancillary Equipment		
3.11	Cable Types and Sizes		
3.12	Heater option		
3.13	Padlockable distribution ways		
3.14	Mechanical cable termination clamps option		

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• <u>IEC 60439-1</u> / <u>BS EN 60439-5</u>				
Clause/Sub-clause		Requirement	Conformance Code	Remarks
IEC 60439-1	BS EN 60439-5			
1.1	1.1	Scope & Object		
2	2	Definitions		
3	3	Classification of Assemblies		
4	4	Electrical Characteristics of Assemblies		
4.1.2		Rated insulation voltage		
4.1.3		Rated impulse withstand voltage		
4.2		Rated current		
4.3		Rated short time withstand current		
4.4		Rated peak withstand current		
4.5		Rated conditional short-circuit current		
4.6		Rated short circuit current		
4.7		Rated diversity factor		
4.8		Rated frequency		
5	5	Information to be given regarding the assembly		
5.1	5.1	Nameplates		
5.2	5.2	Markings		
5.3	5.3	Instructions for installation, operation and maintenance		

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• <u>IEC 60439-1 / BS EN 60439-5</u>				
Clause/Sub-clause		Requirement	Conformance Code	Remarks
IEC 60439-1	BS EN 60439-5			
6	6	Service Conditions		
6.1	6.1	Normal Service Conditions		
6.1.1		Ambient air temperature		
6.1.1.1		Indoor Installations		
6.1.1.2		Outdoor Installations		
6.1.2		Atmospheric conditions		
6.1.2.1		Atmospheric conditions for indoor installations		
6.1.2.2		Atmospheric conditions for outdoor installations		
6.1.2.3		Pollution degree		
6.1.3		Altitude		
6.2	6.2	Special Service Conditions		
	6.2.8	Exposure to heavy vibration and shocks		
6.2.10		Consideration of appropriate remedies against		
6.3		Conditions during transport, storage and erection		
	7	Design & Construction		
	7.1	Mechanical Design		
7.1.1	7.1.1	General		

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• <u>IEC 60439-1 / BS EN 60439-5</u>				
Clause/Sub-clause		Requirement	Conformance Code	Remarks
IEC 60439-1	BS EN 60439-5			
	7.1.2	Clearances, creepage distances and isolating distances		
7.1.2.1		Creepage & Clearance Distances		
7.1.2.2		Isolation of withdrawable parts		
7.1.2.3		Dielectric properties		
7.1.2.3.1		General		
7.1.2.3.2		Impulse Withstand Voltage of the Main Circuits		
7.1.2.3.3		Impulse withstand voltages of auxiliary circuits		
7.1.2.3.4		Clearances		
7.1.2.3.5		Creepage distances		
7.1.2.3.6		Spacings between separate circuits		
	7.1.3	Terminals for external conductors		
7.1.3.1 to 7.1.3.6				
7.1.3.7		Identification of terminals		
7.1.4		Resistance to abnormal heat and fire		

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• <u>IEC 60439-1 / BS EN 60439-5</u>				
Clause/Sub-clause		Requirement	Conformance Code	Remarks
IEC 60439-1	BS EN 60439-5			
7.2	7.2	Enclosure & degree of protection		
7.2.1.1	7.2.1.1			
7.2.1.2	7.2.1.2			
7.2.1.3	7.2.1.3			
7.2.1.4				
7.2.1.5				
7.2.1.6				
7.2.2	7.2.2	Measures To Take Account Of Atmospheric Humidity		
7.3	7.3	Temperature Rise		
7.4	7.4	Protection against electric shock		
7.2	7.2	Enclosure & degree of protection		
7.2.1.1	7.2.1.1			
7.2.1.2	7.2.1.2			
7.2.1.3	7.2.1.3			
7.2.1.4				
7.2.1.5				
7.2.1.6				

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• <u>IEC 60439-1 / BS EN 60439-5</u>				
Clause/Sub-clause		Requirement	Conformance Code	Remarks
IEC 60439-1	BS EN 60439-5			
7.2.2	7.2.2	Measures To Take Account Of Atmospheric Humidity		
7.3	7.3	Temperature Rise		
7.4	7.4	Protection against electric shock		
7.2.2	7.2.2	Measures To Take Account Of Atmospheric Humidity		
7.3	7.3	Temperature Rise		
7.4	7.4	Protection against electric shock		
7.4.2.1.1 & 7.4.2.1.2				
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7.5.3		Relationship between peak current and short-circuit current		
7.5.5.3		Selection and installation of non-protected active conductors to reduce the possibility of short circuits.		
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